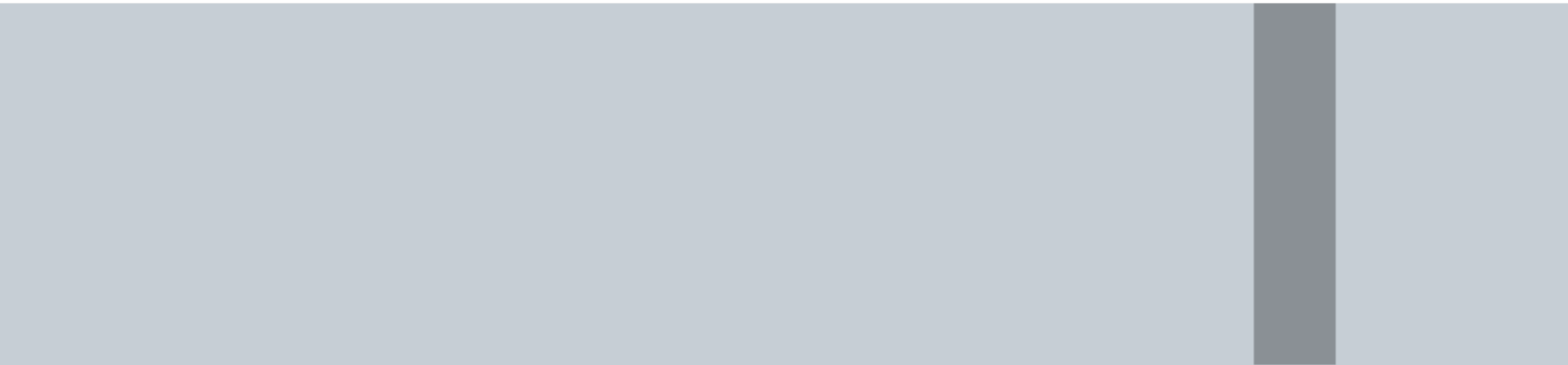
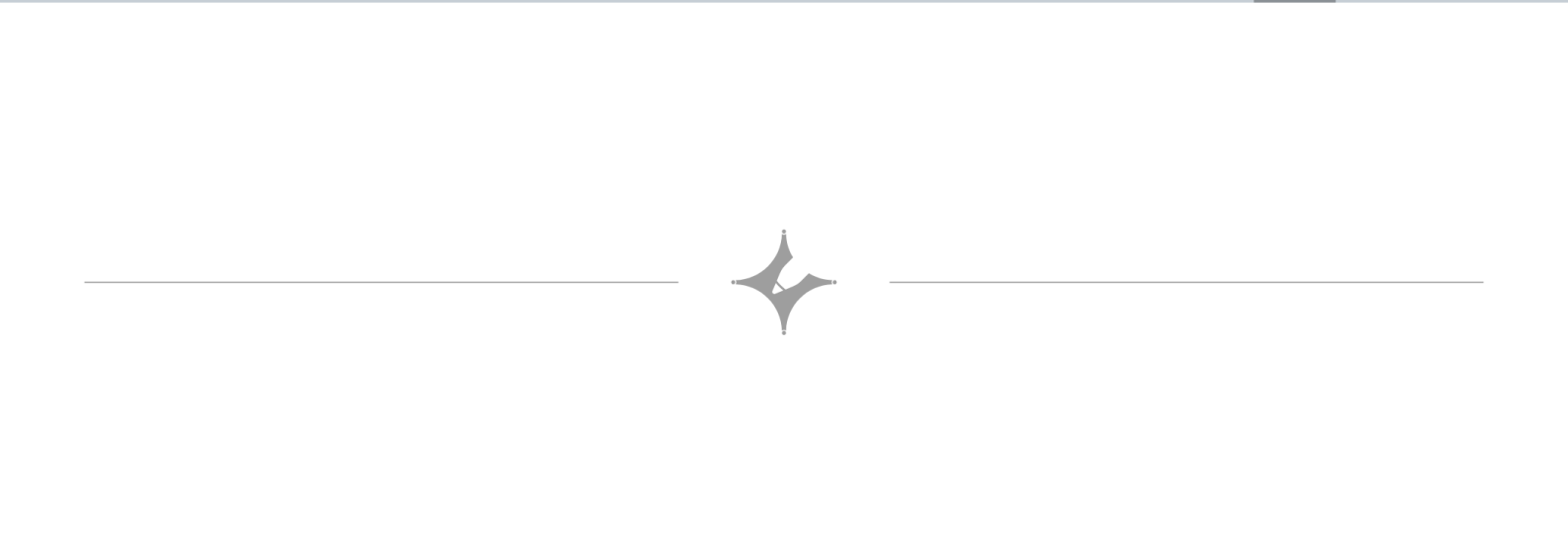
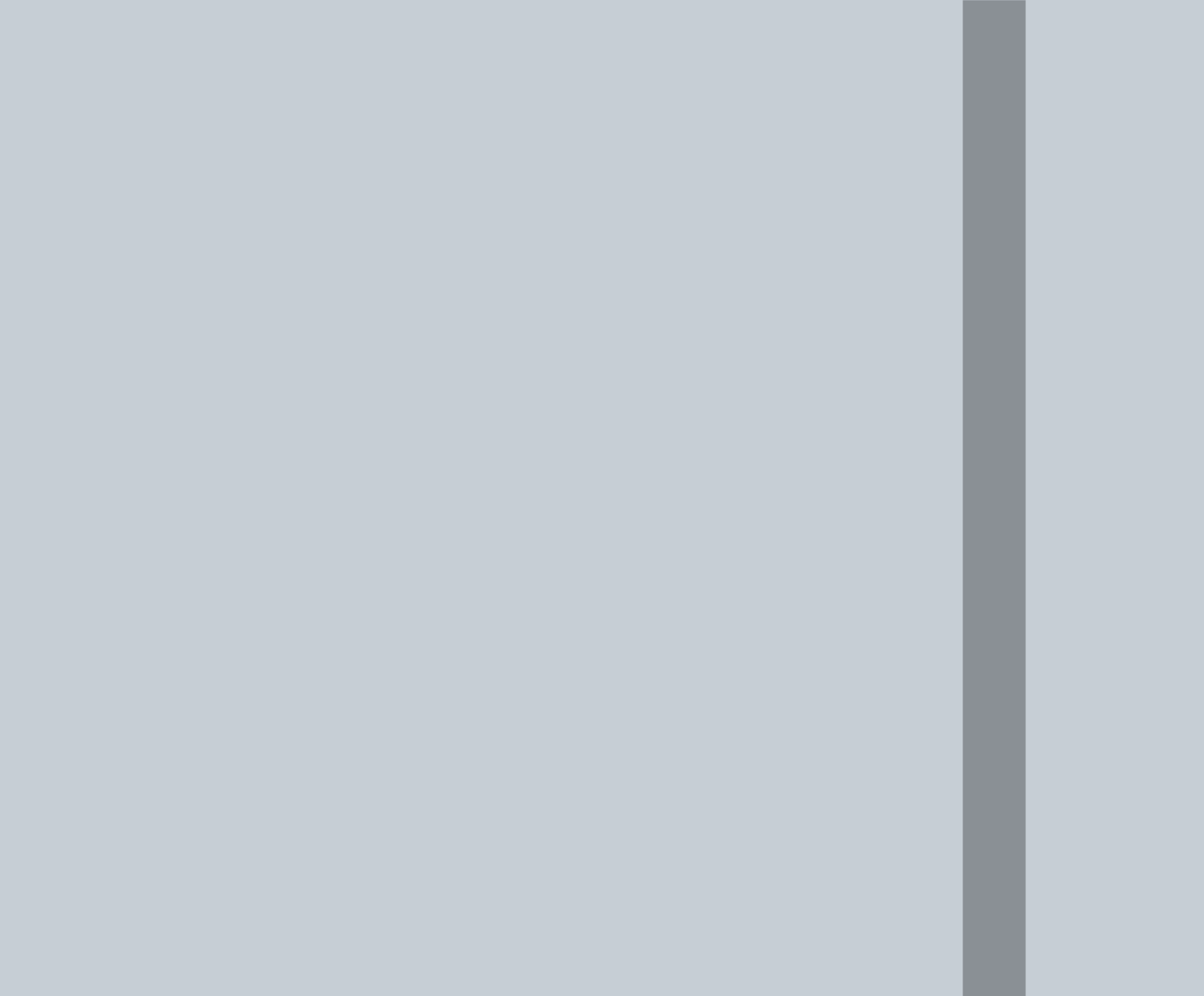




Calendars

(1) Odd days $\rightarrow$ How many days are left after completing a week.
 $\rightarrow$ ex $\rightarrow$ in 10 days $\left(\frac{10}{7}\right) = 3$ days are odd days.
 $\rightarrow$ If today is Monday means 10 days later is Monday + 3 days = Thursday.

(2) In 1 normal yr = $\frac{365}{7}$ = there is 1 odd day.

(3) 1 Leap Yr (LY) has 2 odd days cuz a LY has 366 days.

So, 5 May 2025 $\rightarrow$ Mon
 " " 2026 $\rightarrow$ Tuesday
 " " 2027 $\rightarrow$ Wednesday
 Leap Yr $\leftarrow$ " " 2028 $\rightarrow$ Friday (not Thursday).

(4) If a Yr is divisible by 4 then it's a leap Yr.

(If last 2 digits are divisible by 4 then the entire no. is divisible by 4) (except for centurian Yr).

(5) A centurian Yr (1900, 2000, 2100) has to be divisible by 400 (not 4) for it to become a centurian year.

(6) Months:-

1 Jan $\rightarrow$ Mon	$\rightarrow$ Mon + 3				
1 Feb $\rightarrow$ Thu					
31 J	28 F	31 M	30 A	31 M	30 J
3	0	3	2	3	2
31 J	31 A	30 S	31 O	30 N	31 D
3	3	2	3	2	3

$\rightarrow$ odd days.

1 Jan 2006 } 1 } Sun
07 } 1 }
08 } 2 } 5
09 } 1 }
10 } 1 } Fri

OR

$$4 + 1 = 5$$

$\hookrightarrow 24 - 1 = \frac{23}{7} = 2 + \text{Sun} = \underline{\text{Tuesday}}$

F M A M J J A S
 17 31 30 31 30 31 31 11
 7B 3 3 ~~2 3 2~~ 3 ~~3 4~~
 $\frac{9}{7} \rightarrow 2$ Thu + 2 = Sat

13 Sep 1993 } 1
94 } 1
13 Sep 95 } 2
13 Sep 96 } 1
97 } 1
98 } 1
99 } 2
00 } 2

$\frac{9}{7} = 2 + \text{mon} = \text{wed} + 1$
 $= \underline{\underline{\text{Thu}}}$

* In Q2 we added the leap yr after the yrs but here we added in before simply cuz 29th feb was there in Q1 before but not in Q4.

↳ If 29th feb is there in the time line then take the +2 for leap yr else take +1
(fix this yaad na aaye to chad gpt karle)

$$10 + 1 + 1 = \frac{12}{7} R \rightarrow 5$$

$24 \text{ Apr } 1982 = \text{Mon} + 5 = \text{Sat}$

A	M	J	J	A	S	0	N
6	3	2	3	3	2	3	11

$$\frac{12}{7} \xrightarrow{R} 5$$

11 Nov 1982 = Sat + 5 = Thu

8 Feb 2005

04	2	} 7=0
03	1	
02	1	
01	1	
00	2	

8 Feb Tue

39	1
----	---

2005 - 2000
5 + 1 + 1 = 7 = 0

1999 ka nikadna hata to yaha
 $7+1$ nahi $7-1$ karte in LY
 cuz we are going backwards
 So, 8 Feb 1999 = Monday

8. Which of the following calendars can be reused in 2016?

A. 1999 B. 2004 C. 1988 D. 1994

↳ 2016 = leap yr



8. Which of the following calendars can be reused in 2016?

A. 1999 B. 2004 C. 1988 D. 1994

$$2016 - 2004$$

$$12 + 1 + 1 + 1 = \frac{15}{7} \neq 0 \rightarrow \text{remainder} \neq 0$$

$$2016 - 1988$$

$$28 + 1 + 1 + 1 + 1 + 1 + 1 + 1 \Rightarrow 0 \rightarrow \text{remainder} = 0$$

→ (logic = If a calendar can be reused means from that yr to this yr. The no. of leap days has to be 0.)

10. After 65 days it will be Wednesday. Which day is it today?

A. Sunday B. Wednesday C. Friday D. Monday

$$\rightarrow x + 65 = \text{Wed}$$

$$x = \text{Wed} - 65$$

$$\Rightarrow \frac{65}{7} = 2 \text{ is remainder}$$

$$\text{So, } \text{Wed} - 2 = \underline{\underline{\text{Monday}}}.$$

